Rational certificates for the exact quantum query complexity of nine open cases of $\mathrm{EXACT}^n_{k,l}$

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Abstract

Ambainis, Iraids and Nagaj conjectured that $Q_E(\mathrm{EXACT}^n_{k,l}) = \max\{n-k,l\}-1$ whenever $l-k \geq 2$, where $\mathrm{EXACT}^n_{k,l}(x) = 1$ iff $|x| \in \{k,l\}$. The conjecture is known for $l-k \in \{2,3\}$, for $k=0$, and for $(k,l) = (1,n-1)$. We give exact rational certificates, in the semidefinite characterisation of Barnum, Saks and Szegedy, for nine further instances with $l-k \geq 4$ and $n \leq 9$, which together with the known lower bound determine Q_E in each case and confirm the conjecture there. The certificates are found by a symmetry- and facially-reduced SDP followed by exact rounding, and are checked by a short standalone program. We also report numerical evidence for all 17 open instances with $l-k \geq 4$ and $n \leq 10$.

1 Background

For $f : \{0,1\}^n \to \{0,1\}$ we use the model of [1]: state space $\mathcal{H}_Q \otimes \mathcal{H}_W \otimes \mathcal{H}_O$ with $\mathcal{H}_Q = \mathrm{span}\{|0\rangle, \ldots, |n\rangle\}$, oracle $O_x|i\rangle = (-1)^{x_i}|i\rangle$ for $i \in [n]$, $O_x|0\rangle = |0\rangle$, an algorithm $U_T O_x \cdots O_x U_0$ followed by a measurement whose outcome is read from $\mathcal{H}_O$. $Q_E(f)$ is the least T for which some algorithm outputs $f(x)$ with certainty for every x. Ambainis, Iraids and Nagaj [1] proved

$$\max\{n-k,l\}-1 \;\leq\; Q_E(\mathrm{EXACT}^n_{k,l}) \;\leq\; \max\{n-k,l\}+1, \tag{1}$$

the lower bound by the polynomial method (their Theorem 5) and the upper bound for all $d \geq 1$ (their Theorem 3), and conjectured (Conjecture 1) that the lower bound is tight for $d = l-k \geq 2$, proving it for $d \in \{2,3\}$. Yao and Ye [3] proved it for $k=0$ and for $(k,l) = (1,n-1)$. Wu et al. [4] reported numerical values (no proofs) for even n with k,l symmetric about $n/2$, including $Q_E(\mathrm{EXACT}^8_{2,6}) = 5$. Since $\mathrm{EXACT}^n_{k,l}(x) = \mathrm{EXACT}^n_{n-l,n-k}(\bar{x})$ we list instances with $k \leq n-l$.

The BSS characterisation. Let $E_0 = J$ (all-ones) and $E_i[x,y] = (-1)^{x_i+y_i}$.

Lemma 1 ([2])**.** *There is an exact T-query algorithm for f iff there exist real positive semidefinite matrices $M_t^{(i)}$ $(0 \leq t < T,\ 0 \leq i \leq n)$ and Γ^0, Γ^1, indexed by $\{0,1\}^n$, with $\sum_i M_0^{(i)} = J$; $\sum_i M_t^{(i)} = \sum_i E_i \circ M_{t-1}^{(i)}$ for $1 \leq t < T$; $\Gamma^0 + \Gamma^1 = \sum_i E_i \circ M_{T-1}^{(i)}$; and $\Gamma^b[x,x] = 0$ whenever $f(x) \neq b$.*

*Independent. The search, certificates and code were produced with substantial assistance from an AI system (Claude, Anthropic). All claims are backed by certificates that can be verified independently: `https://github.com/{ORG}/{REPO}`.

Proof of the direction used. Write $M_t^{(i)}[x,y] = \langle v_{t,i}(x), v_{t,i}(y)\rangle$ and $\Gamma^b[x,y] = \langle w_b(x), w_b(y)\rangle$ with real vectors in a common space $\mathbb{R}^r$. Put

$$\psi_t(x) = \sum_i |i\rangle v_{t,i}(x), \qquad \varphi_t(x) = O_x\psi_t(x) = \sum_i (-1)^{x_i}|i\rangle v_{t,i}(x) \quad (x_0 := 0).$$

The first condition says that the unit vectors $\psi_0(x)$ have all pairwise inner products 1, so they coincide and the initial state is independent of x; the others say that $\{\varphi_{t-1}(x)\}_x$ and $\{\psi_t(x)\}_x$ have equal Gram matrices, hence are related by an isometry between their spans, which extends to a unitary U_t; and likewise $\varphi_{T-1}(x) \mapsto |0\rangle w_0(x) + |1\rangle w_1(x)$. Measuring the first register of the final state outputs b with probability $\Gamma^b[x,x]$, which is 0 for $b \neq f(x)$. □

Remark 2. The phase oracle with a null query and the standard XOR oracle simulate each other one query for one (map $|0\rangle \mapsto |1\rangle|+\rangle$, $|i\rangle \mapsto |i\rangle|-\rangle$; conversely conjugate the XOR oracle by H on its target), so the results below hold in either model.

2 Results

Theorem 3. $Q_E(\mathrm{EXACT}^n_{k,l}) = \max\{n-k,l\} - 1$ *for the following* $(n, \{k,l\})$ *(and their complements):*

n	7	8	8	8	9	9	9	9	9
$\{k,l\}$	$\{1,5\}$	$\{1,5\}$	$\{1,6\}$	$\{2,6\}$	$\{2,6\}$	$\{2,7\}$	$\{1,5\}$	$\{1,6\}$	$\{1,7\}$
Q_E	5	6	6	5	6	6	7	7	7

Proof. The lower bound is (1). For the upper bound, the certificate file

`certificates/EXACT_`k`_`l`_n`n`_T`T`.json`

lists rational matrices satisfying the conditions of Lemma 1 with $T = \max\{n-k,l\} - 1$. The program

`certificates/verify_certificate.py`

reconstructs all $2^n \times 2^n$ matrices and checks every condition in exact rational arithmetic, PSD-ness by symmetric Gaussian elimination, and prints the lower bound for comparison. □

Checking takes about 10 s for $n = 7$, minutes for $n = 8$, and much longer for $n = 9$ (the checker is $O(T\,n\,4^n)$ pure Python); SHA-256 hashes of all files are distributed with them. An independent adversarial review (by a separate AI agent, not a human referee) re-derived Lemma 1, wrote a second checker using a characteristic-polynomial PSD test, and confirmed that 31 tampered certificates are rejected.

The matrices are stored compactly. $M_t^{(0)}, \Gamma^b$ are S_n-invariant: an entry $A[x,y]$ depends only on the unordered pair $\{|x|,|y|\}$ and on $|x \wedge y|$. $M_t^{(1)}$ is invariant under permutations fixing coordinate 1, and $M_t^{(i)}[x,y] = M_t^{(1)}[\sigma_i x, \sigma_i y]$ with σ_i the transposition $(1\,i)$. These are not extra assumptions on the verifier's side: it simply rebuilds the full matrices this way and checks them.

3 How the certificates were found

None of this section is needed to check Theorem 3.

Symmetry. Averaging a feasible point over S_n (acting on inputs and on query indices $i \geq 1$) preserves feasibility, so we may assume the invariance above. Invariant matrices block-diagonalise along the $(n-j, j)$ isotypic components of $\mathbb{R}^{\{0,1\}^n}$; we use integer bases built from a Specht-type lowest-weight vector on weight-j strings and the raising operator. For $n = 7$ the blocks have sizes $8, 6, 4, 2$ (S_7) and $14, 10, 6, 2$ (S_6), versus 128.

Facial reduction. Let $V_t = \mathrm{span}\{\chi_S : |S| \leq t\}$. After t queries the amplitudes are polynomials in x of degree $\leq t$, and for an exact algorithm the final amplitudes on output b lie in $W_b = \{h \in V_T : h|_{f^{-1}(1-b)} = 0\}$. A fixed unitary preserves the span of amplitude functions and the oracle multiplies the register-i part by χ_i, so the register-i part before query $t+1$ lies in $F_t^{(i)} = \chi_i Z_{t+1} \cap V_t$, where $Z_T = W_0 + W_1$ and $Z_t = \sum_i F_t^{(i)}$. Hence $\mathrm{range}(M_t^{(i)}) \subseteq F_t^{(i)}$ and $\mathrm{range}(\Gamma^b) \subseteq W_b$. Without the backward part, rounding failed already for PARITY_4.

Rounding. We maximise the least eigenvalue of the face-reduced blocks (in normalised coordinates), round the free variables of the exact RREF of all linear constraints to rationals with denominator 2^{30} or 2^{50}, solve for the pivot variables exactly, and verify positive definiteness on the faces by Sylvester's criterion before the full check. Margins were 0.10–0.17.

Cross-checks. For $(7, \{1, 5\})$ a direct search over unitaries (workspace dimension 3, total dimension 48), polished by Levenberg–Marquardt, gives an explicit algorithm with worst-case error $6 \cdot 10^{-31}$; the certificate itself, converted to unitaries via Lemma 1, has worst-case error $9 \cdot 10^{-16}$. The BSS SDP gives optimal error 0.041 with 4 queries.

4 Numerical evidence

For all 47 instances with $l - k \geq 4$, $6 \leq n \leq 10$, $k \leq n - l$, we solved the symmetry-reduced SDP for the optimal worst-case error ε_T. For each of the 17 open instances, $\varepsilon_{T^*-1} \geq 2.9 \cdot 10^{-4}$ and $\varepsilon_{T^*} \leq 1.0 \cdot 10^{-6}$ with $T^* = \max\{n-k, l\} - 1$. At $n = 10$ the solver accuracy ($\sim 10^{-6}$) approaches the smallest positive values seen ($1.9 \cdot 10^{-6}$ for the proven instance $(10, \{0, 10\})$), so this is weak evidence there.

References

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